Data Profiling Report

ERP Data Cleaning, DB Design & BI Dashboard

Target Dataset: ERP_Uncleaned_Dataset.csv (5,250 records, 27 columns)

1. Executive Summary

This report documents the data profiling audit performed on the raw ERP_Uncleaned_Dataset.csv file, containing 5,250 records across 27 fields spanning 10 operational business entities (Customers, Suppliers, Products, Employees, Sales Orders, Purchase Orders, Inventory, Payments, Shipments, and Returns). The audit identifies every major data quality issue found through a full column-by-column review of the entire dataset (not a sample), states the remediation action taken for each, and explains the reasoning behind decisions where data was deliberately left unresolved rather than fabricated. After cleaning in Power Query, the dataset was reduced from 5,250 to 5,182 rows (68 exact duplicates removed), with all identified formatting, casing, and placeholder issues resolved. Ten major data quality flaws were identified and addressed, summarized in Section 2 below.

2. Data Profiling Audit – Major Data Quality Flaws

The table below lists the ten major data quality flaws identified during profiling, a description of each issue with supporting figures, and the specific cleaning action taken in Power Query.

#	Data Quality Flaw	Description	Cleaning Action Taken
1	Duplicate Records	60 exact duplicate rows initially found; grew to 68 after standardization revealed hidden duplicates masked by casing/spacing differences.	Removed via Power Query "Remove Duplicates." Row count reduced from 5,250 → 5,182.
2	Missing Values (Nulls)	Widespread across the dataset — e.g. Warehouse 42% missing, SupplierName 51%, OrderDate 34%, Quantity 35%, UnitPrice 34%, CustomerEmail 44%.	Categorical/text fields filled with "Unknown." Numeric/date fields (Quantity, UnitPrice, OrderDate) left null intentionally to avoid fabricating financial or transactional data.
3	Text Casing Inconsistencies	CustomerName, SupplierName, EmployeeName, and City contained mixed case (e.g. "BRANDY BARR," "brandon martin") and untrimmed whitespace.	Standardized to Title Case using Power Query's Capitalize Each Word transform, after applying Trim and Clean.

#	Data Quality Flaw	Description	Cleaning Action Taken
4	Inconsistent Geographic Naming	Country field had 10 variant spellings representing only 3 actual countries (e.g. "Pakstan," "PAKISTAN," "pakistan" all meaning Pakistan).	Standardized via Replace Values, then converted to UPPERCASE ISO codes (PK, AE, SA) per assignment specification.
5	Invalid Email Syntax	2,140 of 3,610 non-null emails were malformed. An additional 700 rows contained the literal placeholder text "invalid_email" rather than a real address.	Placeholder text replaced with null. Remaining addresses flagged via a new EmailValidity column (Valid/Invalid) rather than auto-repaired, to avoid inventing data.
6	Placeholder / Fake Data	1,075 Phone entries were the literal value "12345." The ProductName column (1,578 distinct values) was found to be non-representative filler text, not a real product catalog.	"12345" replaced with null. ProductName excluded from the cleaned dataset; replaced with a derived ProductLabel field (Category + ProductID).
7	Logical Errors — Negative Quantities	1,665 rows had negative Quantity values, which is impossible for a sales transaction.	Corrected using absolute value conversion (Number.Abs) in a custom column.
8	Logical Errors — Future Dates	1,690 rows had an OrderDate after the current date, with values as far out as 2027.	Flagged via a new DateFlag column ("Invalid - Future Date") rather than altered, since the true date is unknown and should not be guessed.
9	Pricing Outliers	UnitPrice ranged from 11.13 to 499,619.27. Tested statistically using an IQR (interquartile range) outlier check.	No true statistical outliers found. The wide range was confirmed as legitimate variation across 7 product categories (e.g. Office supplies vs. Electronics/Furniture), not a data error — left unchanged.
10	Referential Integrity Anomalies	ID columns did not map consistently to one entity: 494 of 501 ProductIDs, all 201 SupplierIDs, and all 81 EmployeeIDs were linked to multiple different, unrelated names.	Left unresolved in the raw ID fields by design. New surrogate primary keys will be generated for each entity during 3NF database design in SQL Server (Task 3).

3. Missing Values – Full Column Breakdown

Every column in the dataset was individually profiled using Power Query's Column Quality and Column Distribution views, set to profile the entire dataset rather than the default 1,000-row sample. The table below lists every column with a significant null rate and the handling decision applied.

Column	Null / Missing Count	% of 5,250 Rows	Handling Decision
Warehouse	2,232	42.5%	Filled with "Unknown"
SupplierName	2,684	51.1%	Filled with "Unknown"
CustomerEmail	2,340—(incl. placeholder)	≈ 44%	Placeholder removed; remaining left null, flagged Invalid/Valid
Phone	3,183—(incl. placeholder)	≈ 61%	Placeholder "12345" removed; remainder left as-is (out of scope)
OrderDate	1,772	33.8%	Left null intentionally — no fabricated dates
Quantity	1,851	35.3%	Left null intentionally — not averaged, to protect Total Sales Amount accuracy
UnitPrice	1,829	34.8%	Left null intentionally
CustomerName	1,181	22.5%	Filled with "Unknown"
EmployeeName	1,840	35.0%	Filled with "Unknown"
City / Country	987 / 676	18.8% / 12.9%	Filled with "Unknown"
Category	158	3.0%	Filled with "Unknown"
All 10 ID columns (CustomerID, SupplierID, ProductID, EmployeeID, SalesOrderID, PurchaseOrderID, InventoryID, PaymentID, ShipmentID, ReturnID)	158 each	3.0% each	Left null — fabricating an ID would create false referential links; will become nullable foreign keys in SQL Server

Guiding principle: categorical/label fields (names, cities, statuses) were filled with “Unknown” since this does not distort any calculation. Numeric and date fields tied to financial or transactional accuracy (Quantity, UnitPrice, OrderDate, and all ID/key columns) were deliberately left null rather than filled with an average, zero, or placeholder date — filling over a third of the dataset with a guessed value would have fabricated a large share of downstream figures such as Total Sales Amount, rather than genuinely cleaning it.

4. Refrential Integrity Anamolies

A deeper structural issue was identified beyond simple null/duplicate checks: none of the four primary ID columns reliably map to a single, consistent entity in the raw file.

- CustomerID — every CustomerID with more than one associated row was found linked to multiple different, unrelated customer names (not merely casing variants).
- SupplierID — 201 of 201 distinct SupplierIDs (100%) were linked to more than one distinct SupplierName.
- ProductID — 494 of 501 distinct ProductIDs (98.6%) were linked to more than one distinct ProductName.
- EmployeeID — 81 of 81 distinct EmployeeIDs (100%) were linked to more than one distinct EmployeeName.

Because of this, the raw ID columns cannot be trusted as true keys for building the 3NF relational schema in Task 3. Rather than attempting to “repair” these mappings in Power Query — which would require guessing which name is correct — the decision was made to leave the raw IDs unresolved and generate new surrogate primary keys (SQL Server IDENTITY columns) when building the normalized Customers, Suppliers, Products, and Employees tables.

SalesOrderID shows the same anomaly extending to transactions: a single SalesOrderID was found linked to multiple different CustomerIDs simultaneously (e.g., one order ID appeared across 9 rows with 8 different customers), confirming raw IDs cannot be used to reconstruct real order groupings. This was addressed in Task 3 by generating new surrogate keys rather than trusting source IDs.

Event ID Over-Population:

PurchaseOrderID, PaymentID, ShipmentID, and ReturnID are populated on ~97% of all rows, matching the near-universal fill rate seen across all ID columns — regardless of whether that business event actually occurred (e.g., a ReturnID exists even where no return took place). This means downstream metrics like "return rate" cannot be computed as literal percentages from source ID presence alone; a return rate calculated this way returns ~100%, reflecting this data artifact rather than real return behavior.

5. Place Holder and Non-Representative Data

Two fields were found to contain literal placeholder text rather than genuinely missing or malformed data:

- Phone — 1,075 rows (20.5%) contained the exact literal value '12345,' clearly a placeholder rather than a real phone number. These were converted to null rather than treated as a valid but oddly formatted number.
- CustomerEmail — 700 rows contained the literal text 'invalid_email' as the entire field value. These were converted to null before applying the EmailValidity check, so they would not be miscounted as a distinct email address.

A more significant finding concerned the ProductName column. A full review of all 1,578 distinct values showed the field consisted almost entirely of random, unrelated common English words (e.g. "REGION," "Smile," "Rest," "Develop") rather than real product names, with no dictionary or catalog structure. Because this affects the entire column rather than a subset of rows, no filtering or casing fix could make it usable. ProductName was therefore excluded from the cleaned dataset. In its place, a derived ProductLabel field was created (Category concatenated with ProductID) to provide a usable, non fabricated product identifier for downstream analysis in Tasks 4 and 5.

6. Logical and Numeric Errors

- Negative Quantity: 1,665 rows (31.7%) had a negative Quantity value, which is not physically possible for a sales line item. These were corrected by converting to their absolute value.
- Future OrderDate: 1,690 rows (32.2%) had an OrderDate later than the current date, with values as far forward as July 2027. Rather than altering these to an assumed correct date, a DateFlag column was added marking them "Invalid - Future Date" so they remain visible and excludable in downstream analysis without discarding the original value.
- Zero or negative UnitPrice: checked across all 5,250 rows — none found. This is noted here as a verified clean result, not an oversight.
- Pricing outliers: UnitPrice values were tested using an IQR (interquartile range) statistical method. No values fell outside the calculated outlier fence; the wide observed range (11.13 to 499,619.27) reflects genuine variation across the dataset's seven product categories rather than data corruption.

7. Geographic Naming and Text Casing Standardization

The Country field originally contained 10 distinct spellings representing only 3 actual countries (Pakistan, United Arab Emirates, Saudi Arabia), including case variants ("PAKISTAN," "pakistan"), a misspelling ("Pakstan"), and abbreviation variants ("UAE," "U.A.E," "KSA"). These were consolidated and converted to uppercase ISO-style codes (PK, AE, SA) as specified in the assignment brief.

The City field contained two genuine misspellings in addition to casing inconsistencies: “Qetta” (453 rows) alongside “Quetta” (435 rows), and “Peshwar” (448 rows) alongside “Peshawar” (420 rows). Both were merged into their correctly spelled form. CustomerName, SupplierName, and EmployeeName all had inconsistent casing (e.g. “BRANDY BARR” vs. “brandon martin”) and were standardized to Title Case after trimming leading/trailing whitespace.

City–Country Inconsistency:

Cross-tabulation of the City and Country fields shows no consistent geographic relationship — each city (Karachi, Lahore, Islamabad, Quetta, Peshawar) appears in near-equal proportion across all three countries (Pakistan, UAE, Saudi Arabia), rather than mapping to a single expected country. This suggests the two fields were populated independently rather than reflecting real customer/supplier addresses. Per assignment requirements, Country was standardized to ISO codes (PK/AE/SA) rather than reconciled against City, since City and Country may legitimately refer to different parties in a multi-entity ERP record (e.g., a Pakistani warehouse city shipping to a UAE-based customer's country).

8. Known Limitation

Fulfillment Lead Time:

The assignment brief requests a calculated “Fulfillment Lead Time (Days)” column. This could not be computed: the source dataset contains only an OrderDate field and no corresponding shipment or delivery date field required to calculate elapsed days between order and fulfillment. Rather than approximating this value from ShipmentStatus (e.g. assigning an arbitrary fixed number of days per status), which would fabricate a metric with no basis in the actual data, this field was omitted from the cleaned dataset. This is recorded here as a data availability gap in the source file, not an incomplete cleaning step.

Status Field Distribution:

ShipmentStatus, PaymentStatus, and PaymentMethod each show a near-uniform distribution across their categories (e.g., ShipmentStatus: Completed 24.7%, Processing 24.5%, Cancelled 24.2%, Pending 23.6%). This close-to-random split suggests these fields may not encode a real operational pattern in the source data, and downstream breakdowns by these fields should be interpreted with that caveat.

Inventory Data Absence:

No stock-on-hand quantity or reorder-point threshold exists anywhere in the source data — only an InventoryID label with no accompanying stock figures. Additionally, PurchaseOrders and SalesOrders quantities are drawn from the same underlying row per product (a structural consequence of the flat file combining multiple business events per row), making any "purchased minus sold" inventory estimate mathematically return ~0 rather than a real balance. Inventory valuation and low-stock alerting were therefore omitted from Task 4/5 rather than presented misleadingly.

9. Calculation methodology note

Total Sales Amount was calculated as: Quantity × UnitPrice × (1 − Discount/100), applied only to rows with non-null Quantity and UnitPrice (~42% of rows); no missing values were estimated or averaged.

10. Cleaned Dataset Reflect this Audit

The table below confirms that the cleaned dataset (02_Cleaned_Dataset.csv, exported from Power Query / ERP_Cleaned_Dataset.pbix) reflects every finding and remediation decision documented in this report, verified by re-extracting and re-profiling the final output after cleaning was applied.

Metric	Before Cleaning (Raw)	After Cleaning (Final)
Total Rows	5,250	5,182
Exact Duplicate Rows	60 (68 after standardization)	0
Distinct Country Values	10 (variant spellings)	4 (PK, AE, SA, Unknown)
Distinct City Spellings (Quetta/Peshawar)	Qetta + Quetta, Peshwar + Peshawar (split)	Quetta, Peshawar (merged)
Negative Quantity Rows	1,665	0
Phone = ‘12345’ Placeholder	1,075	0 (converted to null)
Email = ‘invalid_email’ Placeholder	700	0 (converted to null)
ProductName Column	Present (1,578 near-random values)	Removed; replaced by ProductLabel
ID Column Data Type	Decimal (e.g. 1556.0)	Whole Number (Int64)
Quantity Column Data Type	Mixed / Any	Whole Number (Int64)
Total Sales Amount Column Data Type	Not present	Decimal Number, calculated

11. Conclusion

The cleaning process applied to this dataset prioritized transparency over completeness — every value that could be reliably standardized (casing, spelling, geographic naming, negative quantities) was corrected, while values that could not be corrected without guessing (missing dates, missing quantities, broken ID-to-name mappings, unusable ProductName data) were clearly flagged or excluded rather than fabricated. This approach preserves the integrity of downstream calculations such as Total Sales Amount and ensures the SQL Server database design in Task 3 is built on a foundation of known, documented data limitations rather than hidden assumptions.